

Lecture 3 Numbers Theory

Last time: Curry-Howard Correspondence

Lean translates propositions into types.

2 Proof of a proposition corresponds to a term of the corresp. type.

Proof of prop $P \rightsquigarrow p: T(P)$
proof term $\uparrow$ type cons. to P

Upshot: Logical operations are very natural in this setting.

For example implication corresponds to functions!

Today Divisibility & Prime numbers.

Defn Let n, d be nat. numbers, we say

- $d \mid n$ if $\exists k: \mathbb{N}$ s.t. $n = d * k$

6. n is prime if $n > 1 \wedge (d|n \rightarrow d=1 \vee d=n)$

let d, n, m, p be nat. numbers.

Ex. 1) Assume $d|n$ and $d|m$. Show $d|(n+m)$.

2) If p is prime number then p^2 is not
a prime number.

Sol. 1) Choose $k, l \in \mathbb{N}$ s.t. $n = d \cdot k$, $m = d \cdot l$

Then $n+m = dk + dl = d(k+l) \dots$

So using $(k+l)$ makes $d|n+m$.

2) By contradiction assume p^2 is prime.

Observe: $p|p^2$ ✓.

Then since p^2 was assumed to be prime it follows

that $p = 1$ or $p = p^2$.

By assumption on p : $1 < p \Rightarrow 1 < p < p^2$.

So neither of the two hold ~~is~~.

Proof by induction:

Recall: Let $P(n)$ be a prop. about a nat. number n .

Then $P(n)$ holds for all n if

$P(0)$ holds (we can proof $P(0)$).

& $\forall n (P(n) \rightarrow P(n+1))$.

$(P(0) \wedge (\forall n (P(n) \rightarrow P(n+1)))) \rightarrow \forall n P(n)$

Ex Gauss sum: $P(n) : \sum_{k=0}^n k = \frac{n(n+1)}{2}$.

Theorem $P(n)$ holds for all $n \in \mathbb{N}$.

Proof. By induction on n .

• $P(0)$: $\sum_{k=0}^0 k = \frac{0(1)}{2}$
 $\parallel$ $\parallel$
 0 0

• Assume $P(n)$ holds we want to show $P(n+1)$

$$\begin{aligned}\sum_{k=0}^{n+1} k &= \sum_{k=0}^n k + (n+1) \stackrel{P(n)}{=} \frac{n(n+1)}{2} + (n+1) = \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2} \quad \square\end{aligned}$$

Complete / Strong induction

let $P(n)$ be prop. about a nat. number n .

Theorem: $P(n)$ holds for all n if

• $P(0)$ holds

• $\forall n ((\forall m \leq n P(m)) \rightarrow P(n+1))$

(“ P holds $\forall m \leq n$ ”)

Proof. Want to use $Q(n) := \forall m \leq n P(m)$.

[Assume $P(0)$ holds & $\forall n (Q(n) \rightarrow P(n+1))$. Goal: Prove $P(n)$ for all n .

1) $Q(0) \leftrightarrow P(0)$.

3) $\forall n Q(n) \rightarrow \forall n P(n)$

2) $Q(n) \rightarrow P(n)$

4) $(Q(n) \rightarrow P(n+1)) \rightarrow (Q(n) \rightarrow Q(n+1))$

First of all, it suffices to show that $Q(n)$ holds for all n . by 3).

We prove this by induction on n .

- $Q(0) \leftrightarrow P(0)$. by 1) by assumption $P(0)$ holds

- Induction Step: Assume $Q(n)$.

Want to show $Q(n+1)$.

By 4) we know that $Q(n) \rightarrow Q(n+1)$

if $Q(n) \rightarrow P(n+1)$

but that's our assumption from before!

Thus $Q(n+1)$ holds.



Induction on finite sets:

Situation: Indexset I , $|I| < \infty$

Function: $f: I \rightarrow \mathbb{N}$

You want to sum over I .

$\sum_{i \in I} f(i)$ is directly implemented in Lean.

You often want to prove theorems that hold for any such finite set.

You'll use the principle of induction on $|I| = n$.

Simplification: Assume the elements of I live in a type T .

Moreover, we have a proposition $P(I)$ for every finite set I of elts of type T .

Theorem. $P(I)$ holds for all finite sets I .

• If $P(\emptyset)$ holds

• $\forall x:T \quad P(I) \rightarrow P(I \cup \{x\})$
